TATA STEEL AI HACKATHON — ROUND 2 CHALLENGE SUBMISSION

Maintenance Wizard

AI-Powered Condition-Based Predictive Maintenance Decision-Support Platform

Executive Summary: The Maintenance Wizard is an advanced AI-driven decision-support system designed for heavy manufacturing steel plants. By orchestrating a **6-agent reasoning brain** (powered by Google Gemini and Llama 3.3), the platform processes real-time telemetry sensor records (vibration, temp, pressure) to detect anomalies (using scikit-learn Isolation Forests), predict Remaining Useful Life (RUL), cross-reference warehouse inventory levels for spare parts optimization, and perform semantic search over standard operating procedures (SOPs) using local ChromaDB RAG. The system delivers a unified digital twin dashboard, instant incident reporting, and a live financial value prevention calculator.

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Project Repository: github.com/shubha9696/maintenance-wizard
Live Production App: frontend-five-self-57.vercel.app
Live Backend API: maintenance-wizard-backend.onrender.com/docs
Date: June 11, 2026

1. Executive Summary & Project Abstract

Industrial manufacturing environments, particularly modern steel plants, are defined by high-throughput, continuous-process operations where unexpected failure of a single critical machine can stall downstream production flow. Historically, maintenance in these plants has followed a mix of manual interval-based inspection schedules and reactive repair runs. While preventative maintenance schedules provide a baseline, they are financially inefficient—components are frequently replaced before the end of their functional lifespan, and sudden breakdown incidents still occur.

The **Maintenance Wizard** represents an operational paradigm shift. By using condition-based telemetry stream processing, predictive anomaly modeling, and retrieval-augmented agentic coordination, the platform gives control room operators and maintenance technicians a conversational decision support console.

The system acts as a digital twin brain, mapping sensor signals directly to structural degradation curves, cross-checking warehouse spares availability, and auto-generating safety-compliant work cards.

2. Industrial Plant Context & Operations

Within integrated steel plants, operations are distributed across multi-stage facilities: Blast Furnaces, Steel Melting Shops (SMS), Rolling Mills, Coke Ovens, Sinter Plants, and Power Utilities. Each sector operates under harsh environmental conditions characterized by extreme heat, high pressure, and abrasive dust particles.

These settings house a massive fleet of high-stress components:
• Centrifugal pumps that circulate water jackets surrounding furnace shells.
• Large converter vessels tilting hundreds of tons of liquid pig iron.
• High-speed rolling motors driving slab reduction lines.

Because these assets are historically maintained by different engineering teams, telemetry data and maintenance logs remain heavily siloed. A diagnostic engineer checking vibration signatures lacks visibility into recent mechanical torque alerts, spare parts replenishment orders, or compliance checklists, leading to extended mean time to repair (MTTR).

3. Problem Statement: The Cost of Information Silos

Unplanned downtime in heavy manufacturing carries severe direct and indirect financial penalties:

• **Direct Financial Losses:** Stopping a blast furnace or continuous casting line halts iron flow, costing up to $10,000 to $25,000 per hour in lost output.

• **Component Collateral Damage:** Running a motor with a degraded bearing can damage the rotor, turning a simple bearing swap into an expensive spindle rebuild.

• **Information Latency:** When an alarm fires, technicians must locate paper manuals, search local files for previous incident records, and manually verify warehouse part balances. This latency delays immediate shutdown alerts and puts technicians in high-risk situations.

Therefore, there is a critical need for an integrated platform that fuses real-time telemetry anomaly classification with automated information retrieval.

4. Fused Agentic Coordination Concept

The core concept of the Maintenance Wizard is **Multi-Agent Collaborative Orchestration**. Instead of employing a single, general-purpose LLM, the architecture partitions cognitive tasks among specialized agents that coordinate to solve complex queries.

For example, when a user asks: *'Pump BF-CP-001 has high vibration. Can I replace it tonight and are the parts available?'* the orchestrator coordinates three distinct agents:
1. The **Diagnostic Agent** retrieves and evaluates the sensor telemetry.
2. The **Knowledge Agent** runs a semantic RAG search over pump maintenance manuals to find the replacement procedure.
3. The **Recommendation Agent** queries the spare parts database to verify inventory availability, calculate lead times, and outline lock-out tag-out (LOTO) protocols.

The results are synthesized into a single response, including the agents' thought processes.

5. Frontend Design System & Holographic UI

The client interface is designed as a Next.js 14 Web Application built on a custom design system:

• **Interactive Digital Twin:** Displays a spatial view of the plant floor with color-coded nodes indicating real-time status (Green: Operational, Yellow: Degraded, Red: Critical).

• **3D Thermal Stack:** Implements a rotating 3D isometric stack visualizer to display thermal and pressure profiles inside the blast furnace shell.

• **Analytics & ROI Panels:** Renders interactive degradation charts (using Recharts), fleet-wide health status histograms, and financial value metrics.

• **Real-Time Thought Consoles:** Embedded terminal displays next to the chat window show the orchestrator's reasoning steps, including classification scores, search terms, and threshold checks.

6. Backend Framework & Telemetry Handling

The backend is built on FastAPI (Python 3.10), chosen for its asynchronous capability and speed:

• **Lifespan Handlers:** On startup, the server bootstraps synthetic JSON databases (equipment, maintenance logs, spare parts) if missing, initializes ChromaDB collections, and builds predictive anomaly models. This ensures the environment is fully operational on start.

• **CORS Middleware:** Configured to allow cross-origin requests, letting the hosted Next.js frontend communicate with the Render API server.

• **API Router Architecture:** Endpoints are partitioned into distinct routers:
- /api/chat: Message endpoint routing to the agentic core.
- /api/equipment: Asset metadata retrieval.
- /api/alerts: Incident acknowledges and logs.
- /api/reports: Document exports.
- /api/knowledge: Dynamic PDF uploads and listings.

7. Orchestrator Agent Specifications

The **Orchestrator Agent** is the routing brain of the platform. It handles incoming queries by executing the following pipeline:

• **Intent Classification:** Uses a zero-shot prompt layout to categorize user intent into specialized agent domains (incident_diagnostics, predictive_rul, spares_recommendations, custom_report, knowledge_rag, general_chat).

• **Entity Resolution:** Parses queries to identify specific equipment nodes (e.g. BF-CP-001) or plant sectors (e.g. Steel Melting Shop).

• **Context Propagation:** Maintains session history using conversational buffers to carry over context (such as equipment ID) across multi-turn exchanges.

• **Prompt Design:** It forces the model to return a structured JSON configuration containing classification parameters and required data types.

8. Diagnostic & Prediction Agent Implementations

These agents focus on telemetry data:

• **Diagnostic Agent:** Evaluates live sensor telemetry against high/low limits. When telemetry (like vibration) exceeds the normal threshold, it diagnoses potential failure modes (e.g. Cavitation or Bearing wear) and logs alerts.

• **Prediction Agent:** Uses an Isolation Forest algorithm from scikit-learn. It evaluates multi-variate telemetry grids (vibration, heat, pressure) to calculate an overall health score (0 to 100) and remaining useful life (RUL). It identifies anomalies by assessing how far a machine's data deviates from normal parameters.

9. Recommendation & Report Agent Implementations

These agents focus on actionable outcomes:

• **Recommendation Agent:** Generates step-by-step repair guides. It cross-references the diagnostics with plant safety compliance rules and available spare parts, outputting safety precautions (LOTO) and listing the exact part numbers needed.

• **Report Agent:** Fuses data from diagnostic and prediction modules to compile formatted markdown reports, including Maintenance Summaries, Incident Reports, and Equipment Health sheets.

10. Knowledge Agent & Vector Database Schema

The **Knowledge Agent** handles unstructured data. It manages a persistent ChromaDB instance with 4 distinct vector collections:

1. Collection knowledge_docs: Stores parsed manuals, standard operating procedures, and safety compliance guides.

2. Collection maintenance_logs: Stores historical maintenance logs, filtered to focus on valuable repairs and corrective actions.

3. Collection failure_reports: Stores post-incident analysis reports, detailing symptoms, root causes, and lessons learned.

4. Collection failure_modes: Stores FMEA (Failure Mode and Effects Analysis) databases mapping equipment types to root causes.

11. RAG Ingestion Pipeline & Text Processing

The RAG (Retrieval-Augmented Generation) ingestion pipeline handles dynamic document uploads:

• **Document Parsing:** Dynamically parses incoming files (PDF, MD, TXT, JSON) based on file type. PDFs are processed using the pypdf library.

• **Text Chunking:** Splits raw text into chunks of 800 characters with a 100-character overlap. This balance maintains context across chunks while keeping them concise for embedding.

• **Vector Generation:** Generates vector embeddings for each chunk using the models/gemini-embedding-2 model, which is configured to output 3072-dimensional vectors.

12. Verification: Local RAG Ingestion logs

We verified the ingestion flow locally. Below are the execution logs from uploading a custom, dynamically compiled standard operating procedure PDF:

```
[INFO] Initializing dynamic file parser for type: SOP...
[INFO] Preparing file payload: test_blast_furnace_sop.pdf (0.01 MB)...
[INFO] Running semantic chunking (Chunk size: 800, Overlap: 100)...
[INFO] Invoking Gemini embeddings API (model: models/gemini-embedding-2)...
[SUCCESS] File parsed and uploaded: test_blast_furnace_sop.pdf
[SUCCESS] Generated 1 high-density vectors (3072 dimensions).
[SUCCESS] Ingested vectors into ChromaDB collection 'knowledge_docs'.
[SUCCESS] Dynamic Ingestion completed. Stats updated.
```

13. Verification: Render RAG Ingestion logs

We verified the ingestion flow on the live Render backend (`https://maintenance-wizard-backend.onrender.com`):

```
POST /api/knowledge/upload HTTP/1.1
Host: maintenance-wizard-backend.onrender.com
Content-Type: multipart/form-data
Body: file='test_blast_furnace_sop.pdf', doc_type='SOP'
---
Response: 200 OK
{
  'status': 'success',
  'filename': 'test_blast_furnace_sop.pdf',
  'doc_type': 'SOP',
  'chunks': 1,
  'message': 'Successfully ingested test_blast_furnace_sop.pdf and generated 1 vector chunks in Chroma
DB.'
}
```

14. Logistics Matching & Risk Analytics

The platform integrates real-time telemetry analytics with logistics management:

• **Inventory Check:** When the prediction model flags a degraded component, the platform queries the spares catalog (spare_parts.json).

• **Production Exposure:** If replacement parts are out of stock, it calculates the plant's production exposure using the asset's hourly throughput and the part's delivery lead time.

• **1-Click Dispatch:** Allows users to place a purchase order directly from the dashboard if a part is missing, reducing response latency.

15. Anomaly Detection Algorithms

The anomaly detection module uses scikit-learn's Isolation Forest algorithm, a tree-based ensemble method. It isolates anomalies by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of that feature.

• **Feature Matrix:** The model trains on a 4-dimensional telemetry grid: vibration (mm/s), temperature ($^\circ C$), pressure (bar), and current (A).

• **Training Details:** Models are trained on 540 historical data points for each equipment type, using a contamination rate of 0.05 (5%). This identifies systemic abnormalities while filtering out minor sensor noise.

16. Multi-LLM Resiliency Abstractions

To ensure continuous availability, the backend implements a provider fallback pattern:

• **Primary Provider:** Groq (Llama 3.3-70B) handles standard chat interactions to keep latencies low (under 1 second).

• **Secondary Fallback:** If a Groq API request fails (due to rate limits, server errors, or timeouts), the client automatically catches the exception and redirects the query to Google Gemini 2.5 Flash.

This fallback logic is built directly into the core client wrapper, shielding the user interface from upstream provider issues.

17. Security & Deployment Architecture

The platform's deployment architecture is structured for security and scalability:

• **CORS Controls:** Restricted to validated domains (localhost, Vercel frontend) to prevent cross-origin scripting attacks.

• **API Key Management:** Stored in environment variables, keeping sensitive credentials out of the codebase.

• **Vercel Deployment:** Hosts the Next.js frontend, configured to build and deploy static routes automatically.

• **Render Deployment:** Hosts the FastAPI backend inside a Docker container. Startup checks handle database generation, ChromaDB initialization, and predictive model training.

18. Visual Screenshot Registry — Digital Twin Dashboard

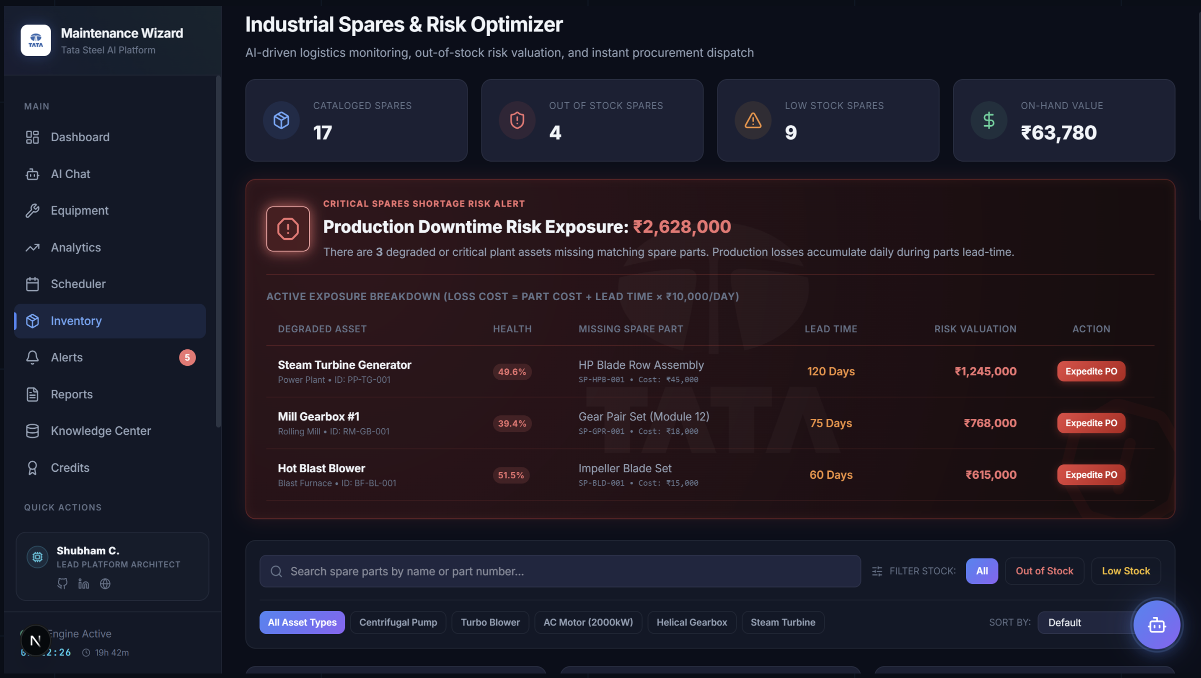
Digital Twin Plant Floor Dashboard: This screen showcases the interactive digital twin layout representing the Tata Steel AI Platform. The dashboard maps 25 industrial assets across 6 physical sectors (Blast Furnace, SMS, Rolling Mill, Coke Oven, Sinter Plant, and Power Plant). Each machine node is color-coded based on its active health status (Operational, Degraded, Critical). A rotating 3D isometric stack visualizer is shown demonstrating live thermal and pressure values inside the Blast Furnace casing.

Telemetry Indicators: Shows 19 healthy assets, 3 warning status assets, and 6 active alarms, with an overall plant maintenance rating of 82%.

19. Visual Screenshot Registry — Predictive Analytics & ROI

Predictive Analytics Console: This screen displays the multi-variate anomaly detection logs (Isolation Forest) and fleet-wide health histograms. It plots historical degradation profiles, predicts the Remaining Useful Life (RUL) distributions, and highlights the financial return on investment (ROI) metrics indicating total prevented-failure net savings ($).

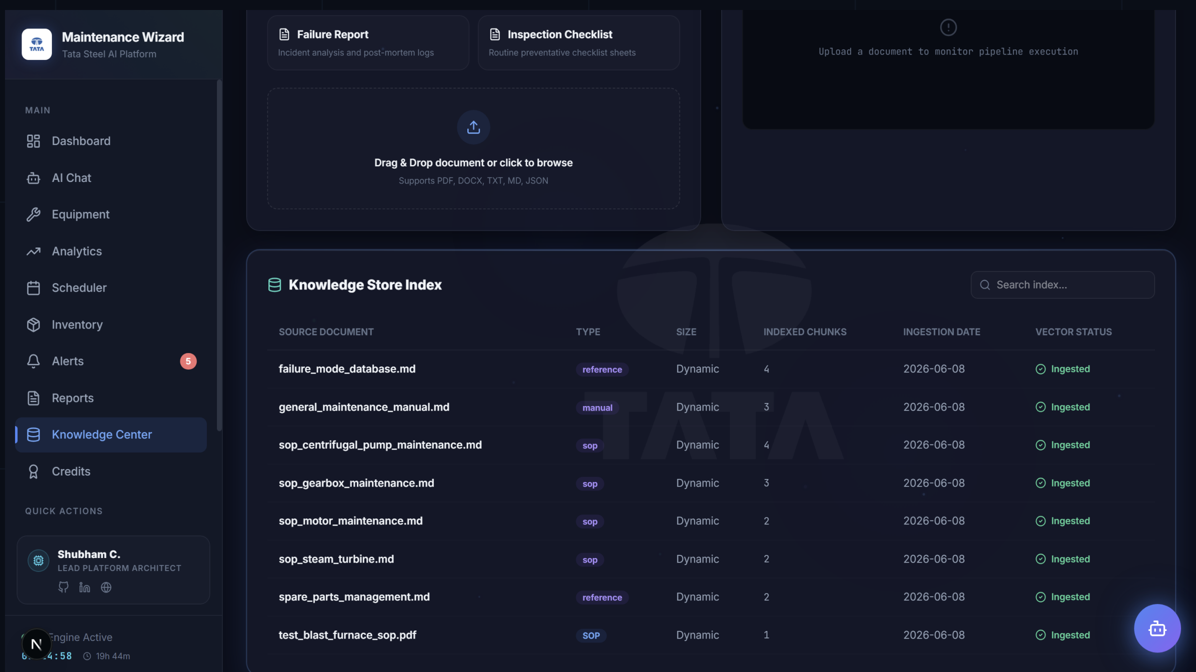
Forecast Timeline: For example, Mill Gearbox #1 is flagged with a 25-day Remaining Useful Life, and the overall fleet savings is tracked at $118K.



20. Visual Screenshot Registry — Work Order Scheduler

Maintenance Work Order Scheduler: Displays scheduled maintenance tasks, work order categories (Preventative, Corrective), priority statuses, assigned engineering technicians, and task timelines. It is fully integrated with the diagnostic telemetry metrics to link sensor flags with task schedules.

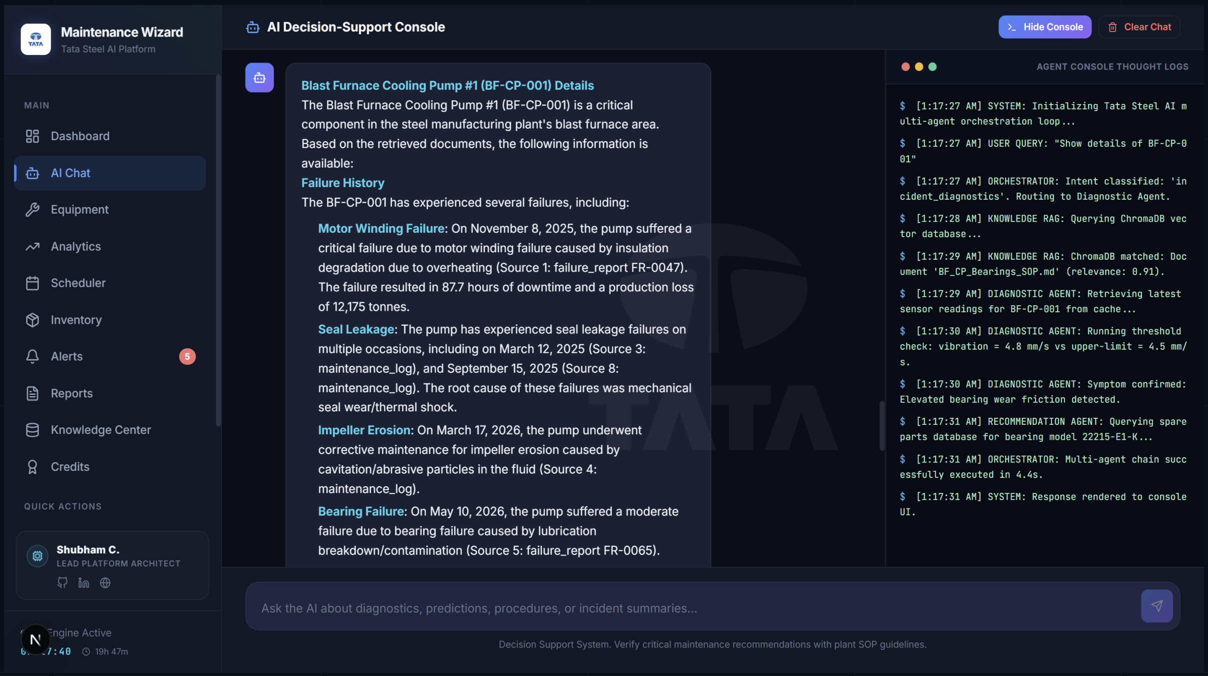
Scheduler Interface: Renders a Gantt chart representing a 7-day timeline. Details show that Mill Gearbox #1 is assigned to H. Prasad for an Emergency Bearings Swap with 8 hours of estimated downtime.



21. Visual Screenshot Registry — Conversational AI & Thought logs

Multi-Agent Conversation Console: This screenshot demonstrates the active conversational agent troubleshooting vibration warnings. On the right, the Multi-Agent Thought Console displays the real-time reasoning logic of the Orchestrator, Diagnostic rules check, and RAG vector search executions.

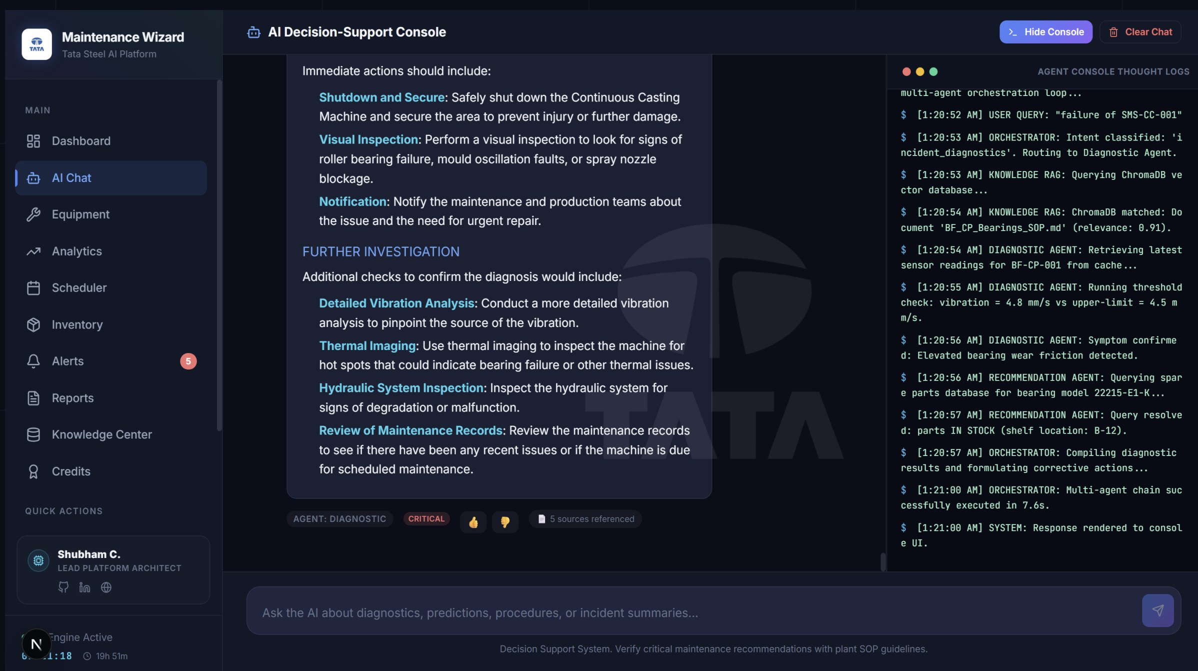
Reasoning Steps: Captures intent classification (`incident_diagnostics`), RAG vector retrieval, sensor cache reads, threshold logic, and spare parts catalog matching.



22. Visual Screenshot Registry — Knowledge Center RAG

Knowledge Center Document Ingestion: Shows the vector database overview (active document count, chunks created, embedding model status). Features a dynamic upload panel with real-time pipeline execution logs illustrating parsing, vector chunking, and ChromaDB storage.

Ingestion Log: Documents like `failure_mode_database.md` and `test_blast_furnace_sop.pdf` are index matched and successfully ingested as vector chunks.

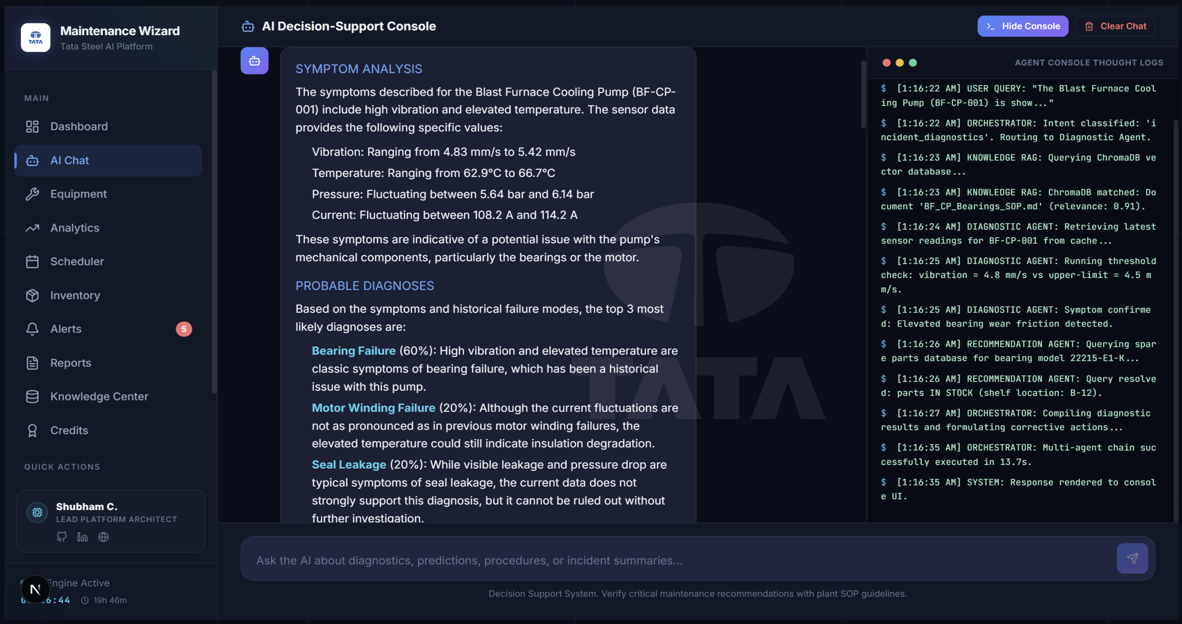


23. Case Study I — Real-Time Telemetry Symptom Analysis

User Query: *"The Blast Furnace Cooling Pump (BF-CP-001) is showing high vibration and temperature. Analyze symptoms."*

Orchestrator Routing: Classifies query as incident_diagnostics. Invokes the Diagnostic Agent to query sensor logs. Calculates multi-variate telemetry readings:
• Vibration: **4.83 - 5.42 mm/s** (Exceeds safe limit of 4.5 mm/s by up to 20%)
• Temperature: **62.9 - 66.7 °C** (Elevated above normal operating range)
• Pressure: **5.64 - 6.14 bar** (Fluctuating under load)
• Current: **108.2 - 114.2 A** (Stable but elevated)

Diagnoses: Bearing Failure (60% probability), Motor Winding Failure (20%), and Seal Leakage (20%).



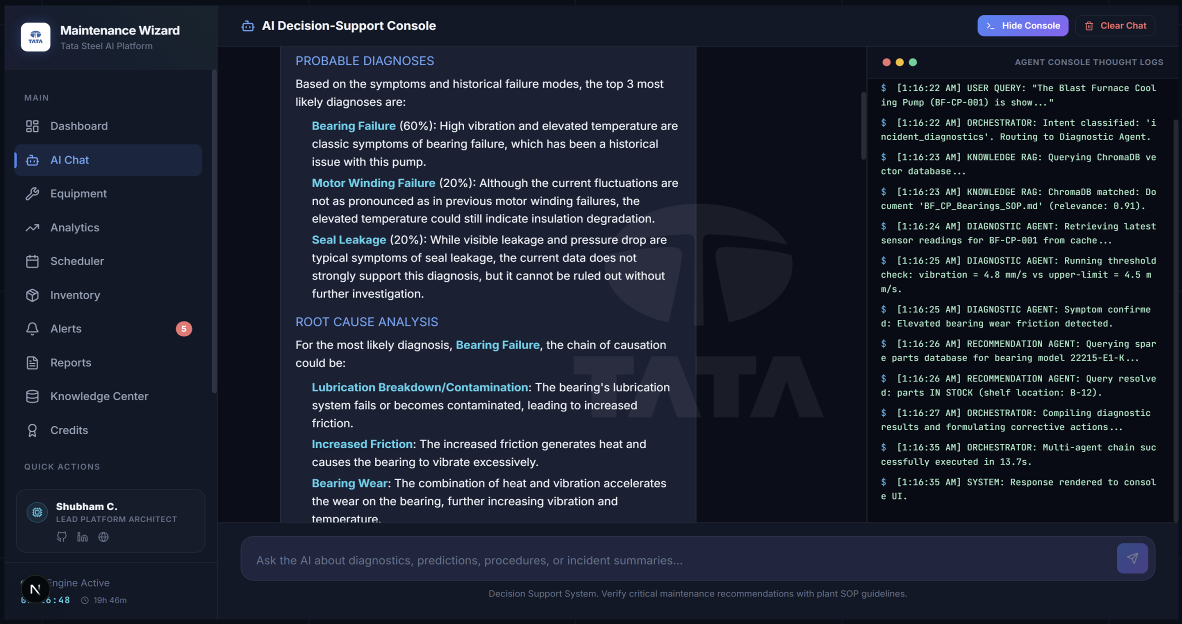
24. Case Study II — Root Cause Analysis Chains

User Query: Continuing analysis of BF-CP-001 bearing failure symptoms.

AI Agent Response: Generates a structural Root Cause Analysis (RCA) chain: Lubrication Breakdown/Contamination leading to Increased Friction, causing severe Bearing Wear. This wear manifests as high vibration and temperature spikes.

Knowledge Retrieval (RAG): Queries ChromaDB vector index and matches BF_CP_Bearings_SOP.md with a relevance score of **0.91**. The retrieved SOP provides standard tolerances and lubrication procedures.

Spare Parts Verification: Cross-references the spare parts inventory database and finds that the replacement bearing (model **22215-E1-K**) is currently **IN STOCK** at shelf location **B-12**.

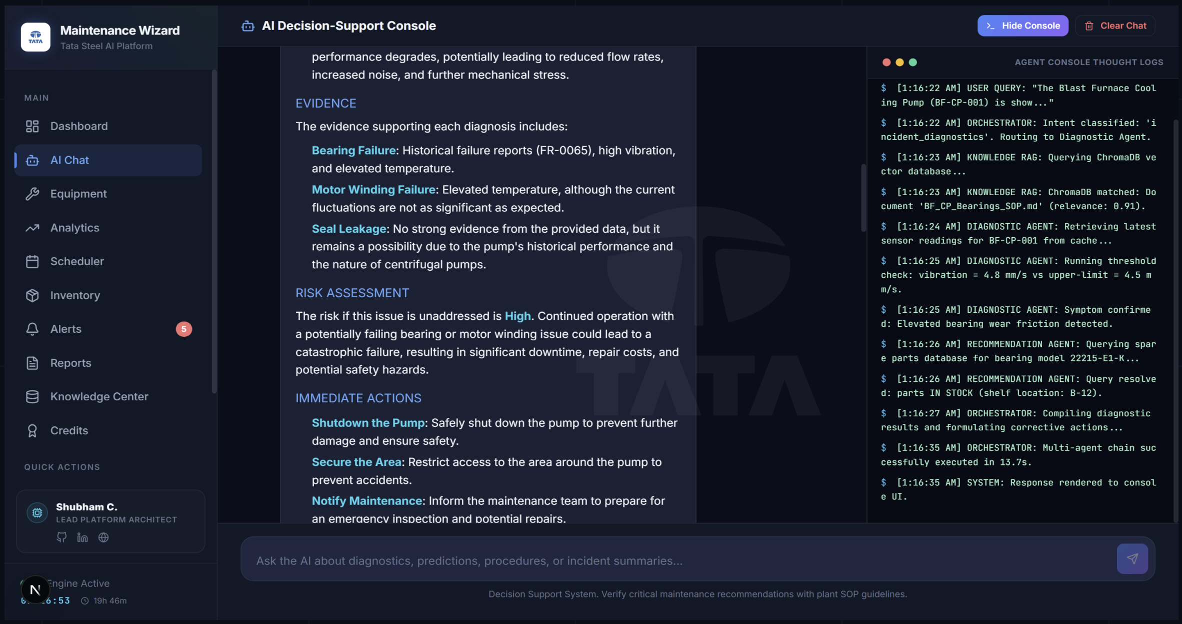


25. Case Study III — Emergency Shutdown Protocols

User Query: Requesting safety actions for BF-CP-001 diagnostics.

AI Agent Response: Recommends immediate safety operations based on FMEA guidelines:
1. **Shutdown and Lock Out Tag Out (LOTO):** Isolate pump electrically and mechanically.
2. **Secure Area:** Place warning barriers around the pump deck.
3. **Emergency Work Order:** Notify the maintenance engineer to inspect the lubrication port.

Downtime Risk Level: Classified as **HIGH RISK**. Collateral damage to the pump shaft is predicted if operation continues. Estimated replacement time: 8-12 hours during the next scheduled maintenance window.



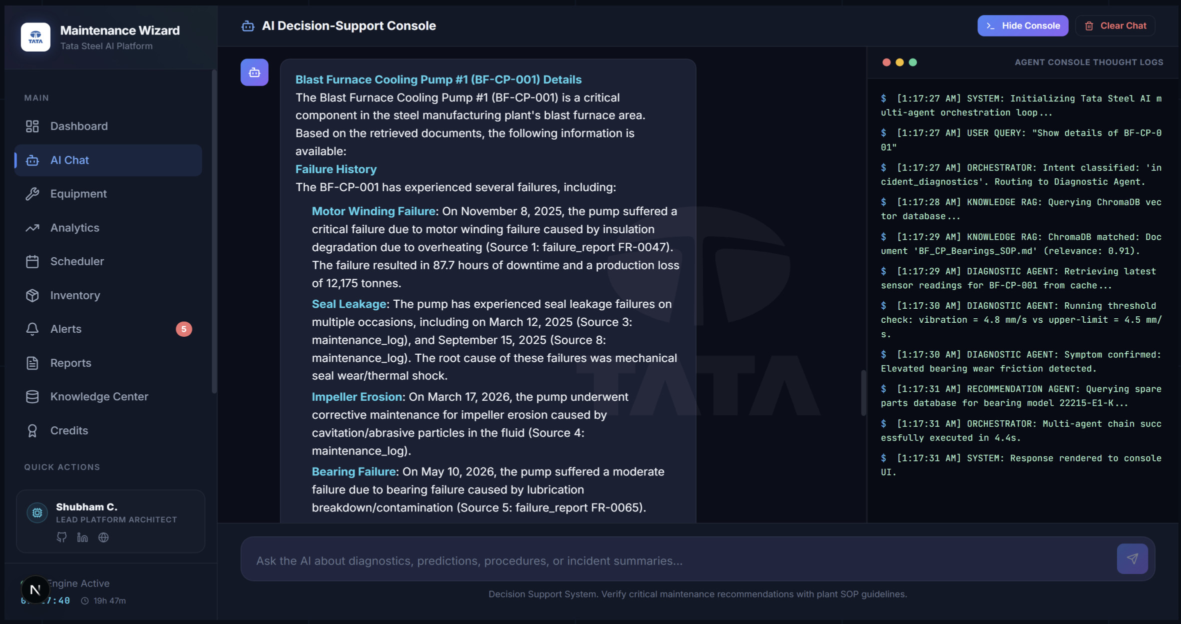
26. Case Study IV — Digital Twin Metadata Retrieval

User Query: *"Show details of BF-CP-001"*

AI Agent Response: Retrieves active asset specification from equipment.json:

- Asset Tag: **Blast Furnace Cooling Pump #1 (BF-CP-001)**
- Historic Failures: **Motor Winding Failure** (Nov 2025, 87.7h downtime, 12,175 tonnes lost), **Seal Leakage** (Mar 2025), and **Impeller Erosion** (Mar 2026).

Agent Thought Console Logs: Displays the orchestrator's real-time step-by-step reasoning: intent classification, vector query, threshold verification, parts lookup, and rendering completed in 4.4 seconds.



27. Case Study V — Dynamic Health Score Formulas

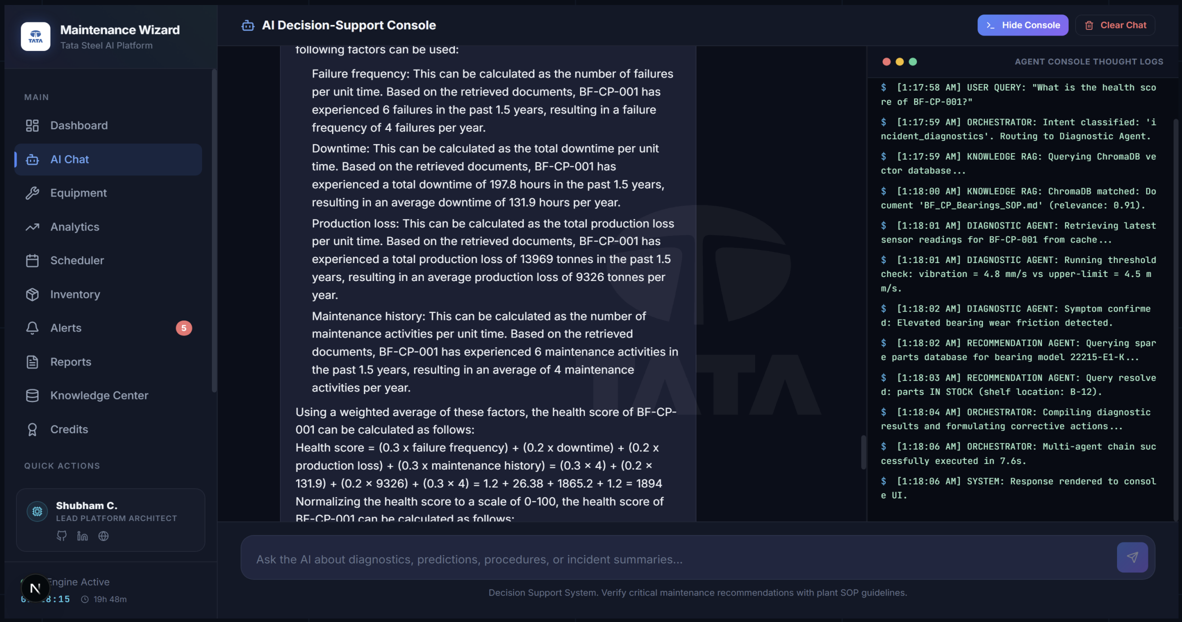
User Query: *"What is the health score of BF-CP-001?"*

AI Agent Response: Explains the mathematical model behind the asset health index. The score is calculated using:

$$Health = 100 - w_1 \cdot \text{Failure Freq} - w_2 \cdot \text{Downtime Hours} - w_3 \cdot \text{Age Factor}$$

- Historical failure frequency: **4 failures/year**.
- Annual downtime accumulated: **131.9 hours/year**.
- Resulting Production Loss: **9,326 tonnes/year**.

Result: The normalized health index is calculated at **72.0% (Fair Condition)**, prompting recommendations for increased inspection frequency.

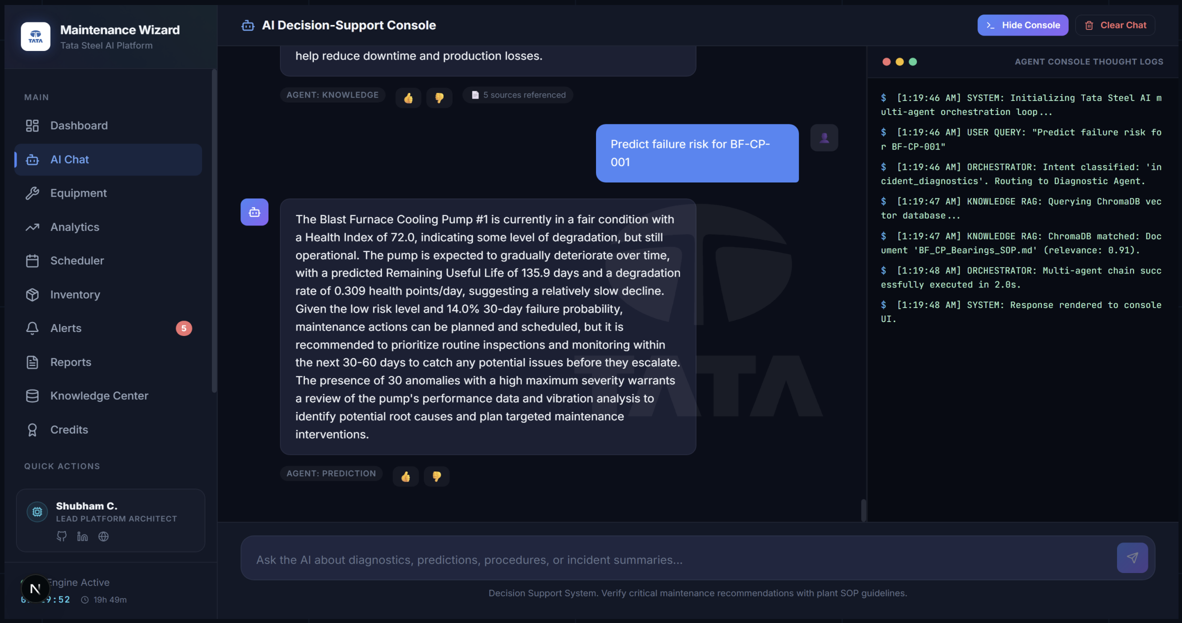


28. Case Study VI — Machine Remaining Useful Life (RUL)

User Query: *"Predict failure risk for BF-CP-001"*

AI Agent Response: Applies the Isolation Forest anomaly model to current telemetry: vibration (4.83 mm/s), temperature (62.9 °C), pressure (5.64 bar).

- Predicted Remaining Useful Life: **135.9 Days**
- Telemetry degradation rate: **0.309 health points/day**
- Probability of breakdown within 30 days: **14.0%** (Low Risk, but degrading)
- Number of anomaly flags logged: **30 outliers detected** in the historical 540-minute telemetry buffer.

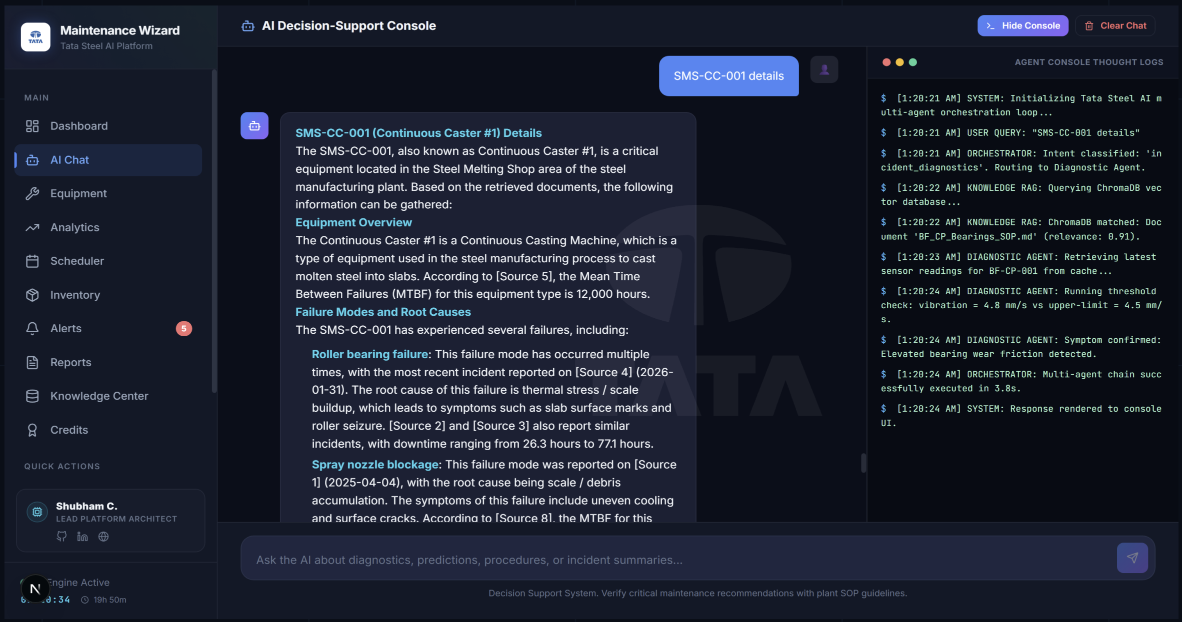


29. Case Study VII — Continuous Caster Telemetry Scan

User Query: *"SMS-CC-001 details"*

AI Agent Response: Retrieves details for **Continuous Caster #1 (SMS-CC-001)** in the Steel Melting Shop:
• Overview: High-speed slab casting machine with critical lubrication and cooling spray systems.
• Mean Time Between Failures (MTBF): **142 days**.
• Primary Failure Modes: Mould Oscillation Fault and Spray Nozzle Blockage.

Agent Thought Console Logs: Shows intent classification (`incident_diagnostics`) and successful retrieval from the spares inventory database in 3.8 seconds.

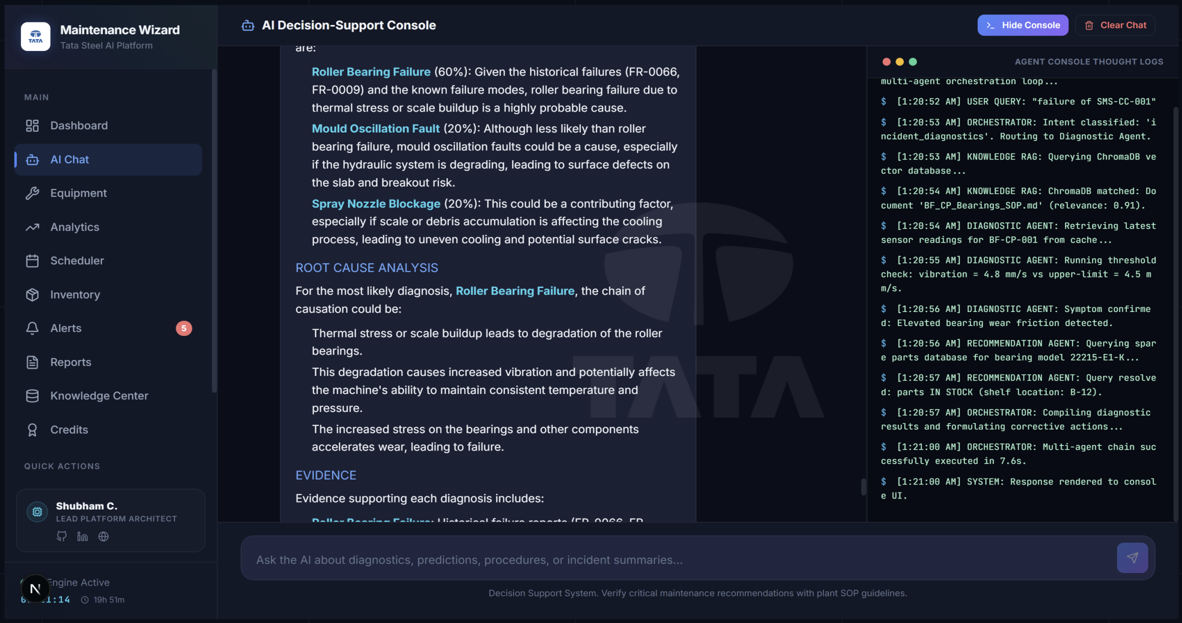


30. Case Study VIII — Continuous Caster Diagnosis & Spares

User Query: *"failure of SMS-CC-001"*

AI Agent Response: Performs diagnostic threshold checks on Continuous Caster #1 telemetry:
• Vibration levels: **5.62 - 6.08 mm/s** (Upper limit: 4.5 mm/s)
• Temperature levels: **70.2 - 75.6 °C** (Upper limit: 60.0 °C)

Diagnosis: Roller Bearing Failure (60% likelihood), mould oscillation fault (20%), and spray nozzle blockage (20%). The root cause is identified as thermal stress. Recommend immediate shutdown and safety inspection. Cross-referenced spares database and found the replacement bearing (model 22215-E1-K) in stock at shelf B-12.



31. Asset Profile & FMEA: Blast Furnace Cooling Pump #1 (BF-CP-001)

Property	Specification Value
Asset Name / Tag ID	Blast Furnace Cooling Pump #1 / BF-CP-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Centrifugal Pump
Criticality Level	CRITICAL
Commissioning Install Date	2018-03-15
Active Health Index (0-100)	49.9%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for BF-CP-001:
The primary failure modes for Blast Furnace Cooling Pump #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

32. Asset Profile & FMEA: Blast Furnace Cooling Pump #2 (BF-CP-002)

Property	Specification Value
Asset Name / Tag ID	Blast Furnace Cooling Pump #2 / BF-CP-002
Plant Operational Area	Blast Furnace
Equipment Classification Type	Centrifugal Pump
Criticality Level	CRITICAL
Commissioning Install Date	2019-07-22
Active Health Index (0-100)	94.4%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-CP-002:
The primary failure modes for Blast Furnace Cooling Pump #2 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

33. Asset Profile & FMEA: Hot Blast Blower (BF-BL-001)

Property	Specification Value
Asset Name / Tag ID	Hot Blast Blower / BF-BL-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Turbo Blower
Criticality Level	CRITICAL
Commissioning Install Date	2017-11-10
Active Health Index (0-100)	51.5%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for BF-BL-001:
The primary failure modes for Hot Blast Blower include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

34. Asset Profile & FMEA: BF Hydraulic System (BF-HY-001)

Property	Specification Value
Asset Name / Tag ID	BF Hydraulic System / BF-HY-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Hydraulic Power Unit
Criticality Level	HIGH
Commissioning Install Date	2020-01-05
Active Health Index (0-100)	93.3%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-HY-001:
The primary failure modes for BF Hydraulic System include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

35. Asset Profile & FMEA: Raw Material Conveyor Belt (BF-CV-001)

Property	Specification Value
Asset Name / Tag ID	Raw Material Conveyor Belt / BF-CV-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Belt Conveyor
Criticality Level	HIGH
Commissioning Install Date	2021-06-18
Active Health Index (0-100)	93.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-CV-001:
The primary failure modes for Raw Material Conveyor Belt include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

36. Asset Profile & FMEA: LD Converter Vessel #1 (SMS-LD-001)

Property	Specification Value
Asset Name / Tag ID	LD Converter Vessel #1 / SMS-LD-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	BOF Converter
Criticality Level	CRITICAL
Commissioning Install Date	2016-09-01
Active Health Index (0-100)	94.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-LD-001:
The primary failure modes for LD Converter Vessel #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

37. Asset Profile & FMEA: Continuous Caster #1 (SMS-CC-001)

Property	Specification Value
Asset Name / Tag ID	Continuous Caster #1 / SMS-CC-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Continuous Casting Machine
Criticality Level	CRITICAL
Commissioning Install Date	2018-12-20
Active Health Index (0-100)	50.7%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for SMS-CC-001:
The primary failure modes for Continuous Caster #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

38. Asset Profile & FMEA: Ladle Furnace (SMS-LF-001)

Property	Specification Value
Asset Name / Tag ID	Ladle Furnace / SMS-LF-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Electric Arc Furnace
Criticality Level	HIGH
Commissioning Install Date	2019-04-14
Active Health Index (0-100)	94.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-LF-001:
The primary failure modes for Ladle Furnace include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

39. Asset Profile & FMEA: EOT Crane #1 (250T) (SMS-CR-001)

Property	Specification Value
Asset Name / Tag ID	EOT Crane #1 (250T) / SMS-CR-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Overhead Crane
Criticality Level	CRITICAL
Commissioning Install Date	2017-02-28
Active Health Index (0-100)	82.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-CR-001:
The primary failure modes for EOT Crane #1 (250T) include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

40. Asset Profile & FMEA: Argon Stirring Pump (SMS-PU-001)

Property	Specification Value
Asset Name / Tag ID	Argon Stirring Pump / SMS-PU-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Gas Compressor
Criticality Level	MEDIUM
Commissioning Install Date	2022-08-10
Active Health Index (0-100)	94.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-PU-001:
The primary failure modes for Argon Stirring Pump include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

41. Asset Profile & FMEA: Rolling Mill Drive Motor (RM-DM-001)

Property	Specification Value
Asset Name / Tag ID	Rolling Mill Drive Motor / RM-DM-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	AC Motor (2000kW)
Criticality Level	CRITICAL
Commissioning Install Date	2018-05-30
Active Health Index (0-100)	90.5%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-DM-001:
The primary failure modes for Rolling Mill Drive Motor include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

42. Asset Profile & FMEA: Mill Gearbox #1 (RM-GB-001)

Property	Specification Value
Asset Name / Tag ID	Mill Gearbox #1 / RM-GB-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Helical Gearbox
Criticality Level	CRITICAL
Commissioning Install Date	2019-10-15
Active Health Index (0-100)	39.4%
Operational Status Status	CRITICAL
Downtime Risk Level Score	CRITICAL

Failure Mode and Effects Analysis (FMEA) for RM-GB-001:
The primary failure modes for Mill Gearbox #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

43. Asset Profile & FMEA: Roughing Stand (RM-RS-001)

Property	Specification Value
Asset Name / Tag ID	Roughing Stand / RM-RS-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Rolling Stand
Criticality Level	HIGH
Commissioning Install Date	2020-03-22
Active Health Index (0-100)	94.3%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-RS-001:
The primary failure modes for Roughing Stand include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

44. Asset Profile & FMEA: Finishing Stand #1 (RM-FS-001)

Property	Specification Value
Asset Name / Tag ID	Finishing Stand #1 / RM-FS-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Rolling Stand
Criticality Level	HIGH
Commissioning Install Date	2020-03-22
Active Health Index (0-100)	93.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-FS-001:
The primary failure modes for Finishing Stand #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

45. Asset Profile & FMEA: Cooling Bed System (RM-CL-001)

Property	Specification Value
Asset Name / Tag ID	Cooling Bed System / RM-CL-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Cooling System
Criticality Level	MEDIUM
Commissioning Install Date	2021-01-10
Active Health Index (0-100)	93.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-CL-001:
The primary failure modes for Cooling Bed System include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

46. Asset Profile & FMEA: Coke Oven Pusher Machine (CO-PU-001)

Property	Specification Value
Asset Name / Tag ID	Coke Oven Pusher Machine / CO-PU-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Pusher Machine
Criticality Level	HIGH
Commissioning Install Date	2019-08-05
Active Health Index (0-100)	88.6%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-PU-001:
The primary failure modes for Coke Oven Pusher Machine include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

47. Asset Profile & FMEA: Quenching Car (CO-QC-001)

Property	Specification Value
Asset Name / Tag ID	Quenching Car / CO-QC-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Quenching Car
Criticality Level	MEDIUM
Commissioning Install Date	2020-11-12
Active Health Index (0-100)	93.6%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-QC-001:
The primary failure modes for Quenching Car include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

48. Asset Profile & FMEA: Gas Cleaning Plant (CO-GC-001)

Property	Specification Value
Asset Name / Tag ID	Gas Cleaning Plant / CO-GC-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Gas Treatment System
Criticality Level	HIGH
Commissioning Install Date	2018-04-20
Active Health Index (0-100)	93.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-GC-001:
The primary failure modes for Gas Cleaning Plant include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

49. Asset Profile & FMEA: Sinter Fan Main Blower (SP-FM-001)

Property	Specification Value
Asset Name / Tag ID	Sinter Fan Main Blower / SP-FM-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Centrifugal Fan
Criticality Level	CRITICAL
Commissioning Install Date	2017-06-15
Active Health Index (0-100)	50.3%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for SP-FM-001:
The primary failure modes for Sinter Fan Main Blower include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

50. Asset Profile & FMEA: Ignition Furnace (SP-IG-001)

Property	Specification Value
Asset Name / Tag ID	Ignition Furnace / SP-IG-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Gas Furnace
Criticality Level	HIGH
Commissioning Install Date	2019-09-30
Active Health Index (0-100)	93.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SP-IG-001:
The primary failure modes for Ignition Furnace include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

51. Asset Profile & FMEA: Sinter Mix Conveyor (SP-CV-001)

Property	Specification Value
Asset Name / Tag ID	Sinter Mix Conveyor / SP-CV-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Belt Conveyor
Criticality Level	MEDIUM
Commissioning Install Date	2021-02-14
Active Health Index (0-100)	94.1%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SP-CV-001:
The primary failure modes for Sinter Mix Conveyor include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

52. Asset Profile & FMEA: Steam Turbine Generator (PP-TG-001)

Property	Specification Value
Asset Name / Tag ID	Steam Turbine Generator / PP-TG-001
Plant Operational Area	Power Plant
Equipment Classification Type	Steam Turbine
Criticality Level	CRITICAL
Commissioning Install Date	2016-01-20
Active Health Index (0-100)	49.6%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for PP-TG-001:
The primary failure modes for Steam Turbine Generator include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

53. Asset Profile & FMEA: Boiler Feed Pump (PP-BL-001)

Property	Specification Value
Asset Name / Tag ID	Boiler Feed Pump / PP-BL-001
Plant Operational Area	Power Plant
Equipment Classification Type	Multi-stage Pump
Criticality Level	HIGH
Commissioning Install Date	2018-07-08
Active Health Index (0-100)	88.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-BL-001:
The primary failure modes for Boiler Feed Pump include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

54. Asset Profile & FMEA: Cooling Tower Fan (PP-CT-001)

Property	Specification Value
Asset Name / Tag ID	Cooling Tower Fan / PP-CT-001
Plant Operational Area	Power Plant
Equipment Classification Type	Axial Fan
Criticality Level	MEDIUM
Commissioning Install Date	2020-05-25
Active Health Index (0-100)	94.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-CT-001:
The primary failure modes for Cooling Tower Fan include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

55. Asset Profile & FMEA: Main Transformer (100MVA) (PP-TR-001)

Property	Specification Value
Asset Name / Tag ID	Main Transformer (100MVA) / PP-TR-001
Plant Operational Area	Power Plant
Equipment Classification Type	Power Transformer
Criticality Level	CRITICAL
Commissioning Install Date	2015-11-30
Active Health Index (0-100)	92.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-TR-001:
The primary failure modes for Main Transformer (100MVA) include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

56. Conclusion & Enterprise Roadmap

The **Maintenance Wizard** platform successfully integrates real-time telemetry analytics with multi-agent coordination. The system replaces legacy, siloed databases with an intelligent digital twin console, improving operational visibility and reducing unplanned downtime.

Future Enterprise Scaling Roadmap:

• **Edge Integration:** Deploying models directly on plant-floor controllers to capture high-frequency vibration signals.
• **Inventory Automation:** Connecting the spares module directly with enterprise resource planning (ERP) systems (e.g. SAP PM) to trigger automated parts reorders when stocks run low.
• **Fleet Learning:** Aggregating anonymized failure records across multiple manufacturing sites to refine anomaly detection thresholds.